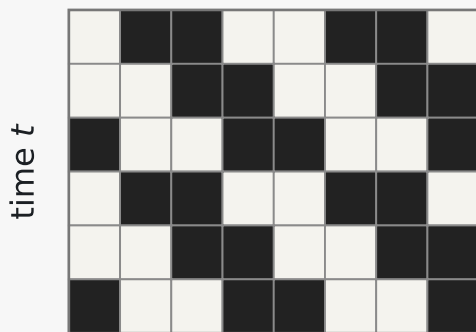


Relative-periodic domain selection by Bernoulli NML

1 Observed spacetime

Choose candidate (p, s) .
Example below: $p = 2, s = 1$.

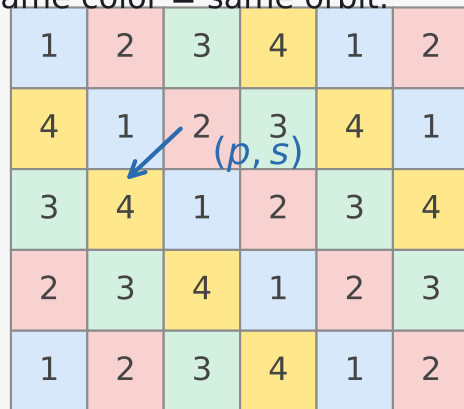
space x



Raw CA spacetime to explain.

2 Orbit classes

Apply $(t, x) \mapsto (t + p, x + s)$.
Same color = same orbit.



Each orbit contributes one background bit.

3 Majority-vote fit

Count ones and zeros in each orbit.
Choose the majority bit.

orbits	ones / zeros
1	5 / 1
2	1 / 5
3	4 / 2

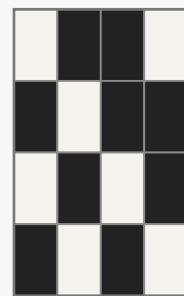
$B^*(p, s)$

Residual mask: $M = U \oplus B^*$.

4 Score and select

NML = fit + complexity.
Keep the best period.

$B^*(p, s)$



$$\text{NLL}(p, s) = \sum_j n_j H_b(\hat{\theta}_j)$$

$$\text{NML} = \text{NLL} + \frac{1}{2} \sum_j \log_2 n_j$$

$$p^* = \arg \min_p \min_s \text{NML}(p, s)$$

Output: periodic background + residual.